

# PERSONAL LEARNING JOURNEY

Member 1 – Kumara D.G.K.R.D. – IT24103972
Assigned role: Problem Framing, Source Validation and Integration

## My contribution

My job was to set up the problem and build the dataset that the whole group used. First, I decided when our prediction is made: after the team sheets are announced, but before the match starts. The model predicts one of three results: home win, draw or away win. Then I found two data sources and combined them into one dataset. The Kaggle European Soccer Database covers 2008 to 2016 and gives FIFA player ratings. Transfermarkt covers 2016 to 2026 and gives player market values. I also kept the group's decision log (19 decisions) and a list of data problems (26 problems, all fixed).

## What I learned

I learned that the moment of prediction decides which data we are allowed to use. Some tables describe what happens during the game, such as goals and cards. If we used them, the model would already be "seeing" the answer, so we left them out. I also learned that an ID number only means something inside its own source. For example, team number 8024 is one club in the first source and a different club in the second. Matching the sources on IDs would have mixed up clubs without showing any error.

## Challenge

The two sources did not fit together easily. They use different team IDs, different player IDs and different ways of measuring player quality. Two leagues, Switzerland and Poland, were missing eight seasons (2016 to 2023), and Poland had no player data for 62.6% of its matches. We also found that one file in our project folder was a wrong version: it differed from the correct file in 28,422 of its 29,606 rows.

## Solution

Instead of matching the sources row by row, I placed one after the other. Each match comes from only one source, and both parts use the same 62 columns. This gave 55,585 matches. After removing Switzerland and Poland, 51,201 matches from nine leagues remained. To make sure both sources describe the same real games, I compared the seasons they share (2012 to 2015). In 99.83% of the matches, the league, date and score were the same. I also saved a digital fingerprint (an MD5 checksum) of each official file, so every member can confirm they are using the right one.

## Reflection

The biggest lesson for me is that many of the risks in a machine learning project are in the data, before any model is trained. One wrong file could have changed every result without anyone noticing. Writing every decision down, with the options and the reason, helped a lot. When someone asked "why did you do this?", the answer was already in the log.

## Future improvement

Next time, I would start the decision log and the file checks on the first day, not later. I would also compare our predictions with bookmakers' odds, which are known before the match, to see how good our model really is. Finally, I would try to link the same players across both sources, so that their history continues after 2016.